Deep Reinforcement Learning-Based Physiological Control for Ventricular Assist Devices

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***Abstract*—With global population aging, the number of patients with end-stage heart failure continues to increase, and ventricular assist devices have become an important treatment option. However, most existing clinical devices still rely on constant pump speed control and cannot dynamically respond to changes in patient activity or resting state, nor can they adequately adapt to inter-patient physiological variability. These limitations reduce physiological perfusion matching and increase the risk of severe complications such as suction, regurgitation, and hemolysis. To address these challenges, this study presents a deep reinforcement learning-based physiological control method for ventricular assist devices. The proposed approach adopts an actor-critic framework with a hybrid CNN-Transformer neural network, where the CNN module extracts local features from physiological signals and the Transformer module captures long-range temporal dependencies. This architecture enables accurate perception of complex hemodynamic states and adaptive pump speed regulation. The proposed method is expected to improve the physiological adaptability, control safety, and hemodynamic performance of ventricular assist devices.**

***Index Terms*—ventricular assist device, physiological control, deep reinforcement learning, actor-critic, CNN-Transformer, hemodynamic regulation**

I. Introduction

Heart failure remains a major global health challenge, and the growing aging population continues to increase the number of patients with end-stage cardiac dysfunction [1]. Ventricular assist devices (VADs) have become an important therapy for these patients because they can provide mechanical circulatory support when the native heart is unable to maintain sufficient perfusion [3]. However, most currently deployed VADs still operate under constant pump speed settings. Although this strategy is simple and reliable, it cannot dynamically match the patient's changing physiological demand during transitions such as rest, exercise, and recovery [2], [3]. As a result, constant-speed control may lead to insufficient physiological perfusion and may increase the risk of adverse events including suction, regurgitation, and hemolysis.

To improve the physiological adaptability of VADs, many studies have explored closed-loop control strategies. Traditional methods, including pressure-based feedback control, fuzzy control, and Starling-like control, can improve performance under specific operating conditions [4]–[7]. Nevertheless, these approaches usually depend on simplified cardiovascular models, handcrafted rules, or manually tuned parameters. Because the cardiovascular system is highly nonlinear, time-varying, and patient-specific, model-based or rule-based controllers often struggle to maintain robust performance under complex hemodynamic disturbances. In addition, control objectives in VAD applications are inherently multi-objective, requiring the controller to simultaneously satisfy perfusion demand, maintain hemodynamic stability, and avoid complications. These requirements make conventional control design increasingly difficult.

Recent advances in artificial intelligence have created new opportunities for physiological control of biomedical devices [8], [9]. In particular, deep reinforcement learning (DRL) is attractive for VAD control because it can learn control policies directly from data without requiring an accurate explicit model of the cardiovascular system. Existing DRL-based studies have demonstrated the potential of reinforcement learning for pump speed regulation [2], [14], [16], but important limitations remain. Some methods rely on multilayer perceptrons or recurrent architectures with limited ability to jointly capture local waveform patterns and long-range temporal dependencies in physiological signals. Other methods focus on a narrow set of reward objectives and therefore do not fully reflect the safety-critical requirements of VAD operation. Consequently, there is still a need for a control framework that combines stronger physiological state representation with a more comprehensive reward design.

To address these issues, this paper proposes a deep reinforcement learning-based physiological control method for ventricular assist devices. The proposed approach adopts an actor-critic framework with a hybrid CNN-Transformer architecture, where the CNN module extracts local features from pressure and flow signals and the Transformer module models long-range temporal dependencies across cardiac cycles. In addition, a multi-objective reward function is designed to encourage adequate physiological perfusion while reducing the risks of suction, regurgitation, and hemolysis. The main contributions of this work are as follows. First, a CNN-Transformer actor-critic control architecture is introduced for VAD physiological control to improve hemodynamic state perception from sequential physiological signals. Second, a multi-objective reward design is developed to integrate physiological regulation and complication avoidance into a unified learning framework. Third, the proposed method provides a foundation for more adaptive, safe, and intelligent pump speed regulation in next-generation VAD systems.

II. Related Work

Existing VAD control strategies have evolved from constant-speed operation to more advanced physiological control schemes [3]. Conventional closed-loop methods, including pressure-based feedback control, proportional-integral regulation, fuzzy control, and Starling-like control, have shown the ability to improve pump response under specific operating conditions [4]–[7]. These methods are attractive because of their conceptual simplicity and clear control objectives. However, they typically depend on simplified cardiovascular models, manually selected control indicators, or extensive parameter tuning. As a result, their performance may degrade when confronted with the highly nonlinear, time-varying, and patient-specific nature of cardiovascular dynamics. In addition, traditional approaches often optimize a limited set of variables and therefore face difficulties when multiple safety and perfusion objectives must be satisfied simultaneously.

With the development of artificial intelligence, learning-based methods have been introduced to improve state perception and decision making for VAD regulation. Some studies have used recurrent neural networks to classify abnormal and normal physiological states, which helps enhance physiological signal interpretation but is still closer to state recognition than to full closed-loop control [8]. Other works have incorporated neural networks into existing control architectures to compensate for uncertainties or estimate hidden hemodynamic variables [10]. Convolutional neural networks have also been explored for sensorless estimation tasks, such as preload estimation from pump flow waveforms, demonstrating the potential of deep models to extract informative features from physiological signals [12]. Nevertheless, many of these approaches focus on diagnosis, estimation, or auxiliary modeling rather than directly learning an adaptive physiological control policy.

Deep reinforcement learning has further advanced the development of intelligent VAD controllers by offering a data-driven framework for continuous decision making. Recent studies based on PPO, DDPG, and TD3 have shown that reinforcement learning can regulate pump speed according to changing physiological conditions and can improve the adaptability of VAD operation [2], [14], [16]. Despite this progress, important limitations remain. In several existing studies, the actor and critic networks are built on multilayer perceptrons or recurrent modules that cannot simultaneously capture local waveform variations and long-range temporal dependencies with sufficient effectiveness [2], [14]. Moreover, some reward designs emphasize only a narrow subset of performance objectives, such as average flow or selected hemodynamic indicators, and therefore do not fully address the safety-critical requirements of VAD control, including the prevention of suction, regurgitation, and hemolysis.

Motivated by these gaps, this work aims to improve both physiological state representation and control objective design in deep reinforcement learning-based VAD regulation. The proposed method combines a CNN-Transformer architecture with an actor-critic framework to jointly model local signal patterns and long-term temporal information. In addition, a multi-objective reward function is introduced to better integrate physiological perfusion demand with safety constraints. In this sense, the present work extends existing research from general intelligent control toward a more comprehensive physiological control framework for ventricular assist devices.

III. Methods

This section presents the proposed deep reinforcement learning-based physiological control framework for ventricular assist devices. The overall method formulates VAD regulation as a continuous control problem and combines an actor-critic learning strategy with a CNN-Transformer state representation model and a multi-objective reward mechanism.

A. Problem Formulation

The VAD physiological control task is formulated as a sequential decision-making problem. At each control step, the controller receives a state vector composed of measured or estimated physiological signals, such as pressure- and flow-related waveforms and their recent temporal evolution. Based on this information, the controller outputs a continuous pump speed command to regulate the mechanical support provided by the device. The control objective is not only to maintain adequate perfusion under changing physiological demand, but also to preserve hemodynamic stability and reduce the risk of adverse events.

From a reinforcement learning perspective, the physiological state of the cardiovascular system defines the environment state, the pump speed adjustment corresponds to the action, and the immediate control quality is reflected by a reward signal. Because VAD regulation must continuously adapt to nonlinear and patient-specific dynamics, the control policy is required to learn from sequential interactions rather than from a fixed analytical model. This formulation allows the proposed method to address dynamic transitions such as rest-to-exercise changes and other time-varying physiological conditions in a unified framework.

B. Actor-Critic Framework

The proposed controller is built on an actor-critic reinforcement learning framework, which is well suited for continuous control problems such as pump speed regulation. The actor network maps the current physiological state to a control action, while the critic network evaluates the long-term utility of the policy by estimating the expected return under the current state-action relationship. Through iterative interaction with the environment, the actor is updated to generate more effective physiological control actions, and the critic provides guidance to stabilize policy learning. This design follows the general line of recent DRL-based VAD control studies while extending them with a stronger state representation module [2], [14], [16].

Compared with rule-based or purely supervised approaches, the actor-critic framework offers two main advantages in this problem. First, it can directly optimize sequential control performance under long-term physiological objectives instead of relying on manually designed control laws. Second, it can naturally support adaptive decision making in the presence of nonlinear cardiovascular dynamics and operating condition changes. This property is particularly important for VAD control, where the controller must continuously balance support effectiveness and safety under uncertain physiological disturbances.

C. CNN-Transformer Architecture

To improve physiological state representation, a hybrid CNN-Transformer architecture is adopted in the actor-critic framework. The CNN module is responsible for extracting local morphological features from physiological waveforms. This design is beneficial for identifying short-term signal variations and waveform patterns that may reflect instantaneous hemodynamic changes or early signs of unsafe operating conditions. In the VAD setting, such local information is important because adverse events may first appear as subtle disturbances in pressure or flow signals. The use of convolutional representations is also motivated by prior sensorless estimation studies that showed CNN-based models can effectively capture informative waveform features in VAD applications [12].

The Transformer module is introduced to capture long-range temporal dependencies in sequential physiological data. Unlike architectures that mainly focus on local or short-memory representations, the Transformer can model interactions across multiple cardiac cycles and therefore provide a more comprehensive description of evolving hemodynamic states. By combining CNN-based local feature extraction with Transformer-based temporal modeling, the proposed architecture is designed to improve physiological state perception and provide richer representations for both the actor and critic networks. This design is intended to address the representational limitations observed in previous DRL controllers that mainly relied on multilayer perceptrons or recurrent structures [2], [14]–[16].

D. Reward Design

A multi-objective reward function is designed to incorporate both physiological regulation and safety constraints into policy learning. The reward encourages the controller to maintain physiologically appropriate perfusion and stable hemodynamic behavior under changing operating conditions. At the same time, it penalizes unsafe or undesirable states associated with suction, regurgitation, and hemolysis, so that the learned policy does not improve one objective at the expense of clinical safety. This formulation extends prior DRL-based VAD reward designs that often emphasized a narrower subset of control objectives [2], [14], [16].

This reward design reflects the multi-objective nature of VAD physiological control. In practice, maintaining a single indicator within a desirable range is insufficient, because effective support must simultaneously account for perfusion demand, cardiovascular stability, and complication avoidance. Therefore, the proposed reward mechanism translates clinical and physiological considerations into learning signals that guide the controller toward safer and more adaptive behavior. By embedding these constraints into the reinforcement learning objective, the method aims to achieve a better balance between support performance and operational safety.

IV. Experimental Setup

The experimental study is designed to evaluate the proposed controller under physiologically relevant operating conditions and to examine its adaptability, robustness, and safety. Following the overall technical route of this work, the experimental setup combines a controllable simulation or mock circulatory loop environment with staged model development, including offline pretraining and online reinforcement learning.

A. Platform and Physiological Scenarios

The evaluation platform is constructed to reproduce the dynamic behavior of the cardiovascular system under different physiological conditions. A simulation environment or mock circulatory loop is configured to emulate vascular resistance, compliance, and other fluid dynamic characteristics relevant to VAD operation. By adjusting platform parameters, the experiments are designed to represent different degrees of heart failure and different activity levels, thereby exposing the controller to a wide range of hemodynamic states. Compared with purely numerical simulation, the mock circulatory loop setting also provides a more realistic environment with flow inertia, sensor noise, and physical disturbances.

B. Data Collection and Training Procedure

The training pipeline follows a staged strategy. First, data are collected under multiple operating conditions by manually adjusting the experimental platform parameters and recording representative physiological signals. These data are used for offline model initialization so that the CNN-Transformer architecture can acquire preliminary feature extraction and physiological state representation capability before closed-loop policy learning. After this stage, the controller is further optimized through online reinforcement learning by interacting directly with the simulation or physical test environment. This design aims to improve training stability and to reduce the difficulty of learning from scratch in a highly nonlinear control problem.

C. Baseline and Evaluation Protocol

The proposed method is evaluated against representative conventional and learning-based control strategies. In addition to overall control performance, the experiments are designed to include comparative validation and ablation analysis so that the contributions of the CNN and Transformer components can be separately examined. The evaluation focuses on physiological adaptability, hemodynamic regulation quality, and safety-related behavior. Special attention is given to the controller response under challenging scenarios, such as sensor disturbance or sudden physiological abnormalities, in order to assess whether the learned policy can maintain stable support while reducing the risks of suction, regurgitation, and hemolysis.

V. Results and Discussion

The proposed controller is evaluated under different physiological scenarios to examine its adaptability, regulation quality, and safety-related behavior. The results are intended to show whether the learned policy can adjust pump speed according to changing hemodynamic demand while maintaining stable support. In comparative experiments, the proposed method is analyzed against representative baseline controllers to assess the benefits of deep reinforcement learning-based physiological control. Particular attention is given to control behavior during transitions across operating conditions, because these situations are critical for evaluating whether the controller can move beyond the limitations of constant-speed or weakly adaptive regulation strategies.

From a physiological regulation perspective, the most important observation is whether the controller can achieve a better balance between perfusion support and hemodynamic stability. A desirable result is that the proposed method maintains more appropriate support across different activity levels and heart failure conditions, rather than optimizing a single variable at the expense of overall cardiovascular behavior. In addition to average regulation quality, the temporal smoothness and responsiveness of pump speed adaptation are also important discussion points, since excessive oscillation or delayed adjustment may reduce control effectiveness and increase clinical risk.

The discussion also focuses on safety-related performance. Because VAD control is a safety-critical task, the evaluation should consider the controller response when the system is exposed to challenging disturbances, such as sensor perturbation or abrupt physiological abnormalities. Under these conditions, the proposed method is expected to show improved robustness and a lower tendency to enter unsafe regimes associated with suction, regurgitation, or hemolysis. If such behavior is confirmed, it would indicate that the multi-objective reward mechanism successfully guides policy learning toward safer operating strategies.

Finally, the ablation analysis provides insight into the contribution of each major module in the proposed framework. By comparing variants with and without the CNN or Transformer components, the experiments can reveal whether local waveform feature extraction and long-range temporal modeling both play meaningful roles in physiological state representation. This analysis is important because it connects the observed control performance back to the main design choices of the method and clarifies why the proposed CNN-Transformer actor-critic architecture is suitable for VAD physiological control.

VI. Conclusion

This paper presents a deep reinforcement learning-based physiological control method for ventricular assist devices. To address the limitations of conventional control strategies and existing learning-based approaches, the proposed method combines an actor-critic reinforcement learning framework with a CNN-Transformer architecture for physiological state representation and a multi-objective reward design for integrating perfusion support and safety constraints. In this way, the method is intended to improve the adaptability of pump speed regulation under nonlinear and time-varying cardiovascular conditions.

The overall study is organized around a staged experimental framework that includes physiologically relevant platform construction, offline model initialization, and online policy learning. The corresponding evaluation is designed to examine physiological adaptability, hemodynamic regulation quality, robustness to disturbances, and safety-related behavior. Together, these components provide a systematic basis for assessing whether the proposed controller can support more effective and safer VAD regulation than existing approaches.

In summary, this work contributes a more comprehensive intelligent control framework for VAD physiological regulation by jointly improving state representation and reward design. Future work will focus on completing extensive experimental validation, refining the controller under more complex pathological scenarios, and advancing deployment on embedded hardware and physical platforms for practical biomedical applications.

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